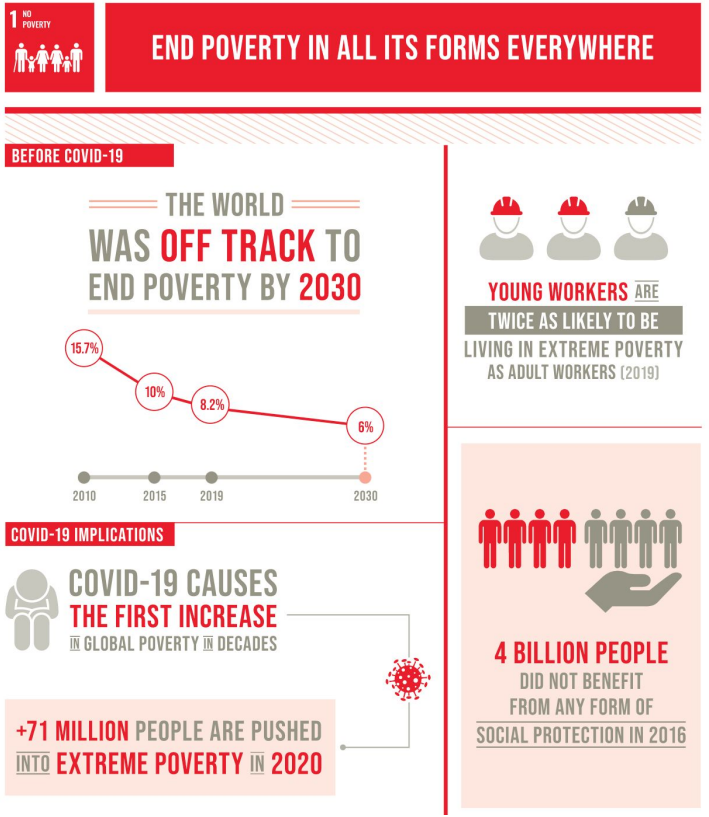
Creating Infographics with Canva

Digital Integration Teaching Initiative
Fall 2026
Dr. Sembe: ENGW 3307
Advanced Writing in the Science Professions
DITI Consultants: Sayyara Poliferno & Tiffany Cruz

Workshop Agenda

- The most common visualizations
- Learn about dynamic and static modes of conveying information.
- Discuss best practices for visualizing data.
- Explore how to make visualizations using Canva, a free online tool.
- Consider accessibility in designing visualizations

Slides and handouts available at:
https://bit.ly/fa26-sembe-engw3307-canva

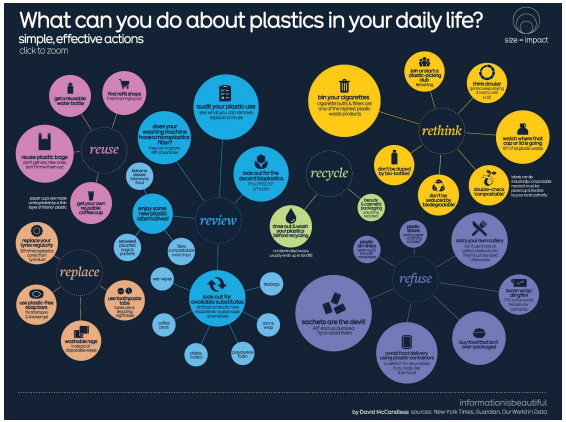


SDG Report 2020

The most common visualizations (1/2)

Infographics - complex information quickly and clearly

Tables - organize data into a grid structure using rows and columns



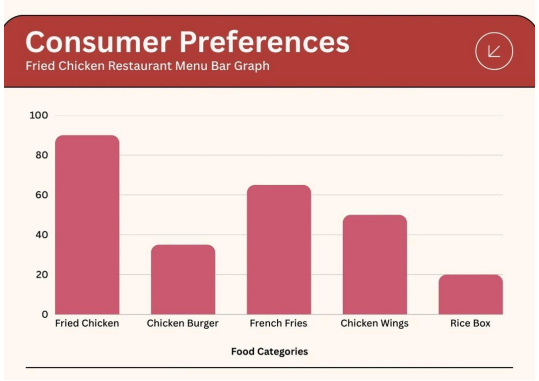
BORCELLE FITNESS

COMPARISON CHART
REALLYGREATSITE.COM

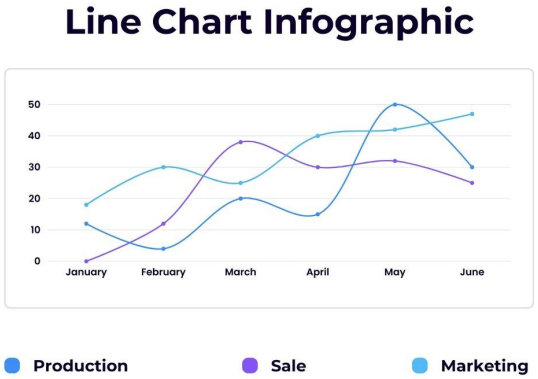
FEATURES	FREE	STANDAR	PREMIUM	VIP
AERIAL YOGA	✓	✓	✓	✓
GROUP EXERCISE	✓	✓	✓	✓
LOUNGE AREA			✓	✓
LOCKERS	✓	✓	✓	✓
SWIMMING POOL		✓	✓	✓
PILATES				✓

The most common visualizations (2/2)

Bar charts - sort data from highest to lowers, comparing quantitative data across different categories



Line graphs - show time related data, fluctuations, and trends



Infographics

Infographics Basics 1/2

- Infographics present complex information quickly and clearly.
- Infographics tell a story with **structure** and **sequence**, and are most often images combined with numbers, charts, graphs, and summary text.
- Infographics **concisely** convey information and data to different audiences **across platforms and media** (i.e., digital or print).

Feel free to ask questions at any point during the presentation!

Infographics Basics 2/2

- Infographics have **narrow** focus or subject matter, **short** and accessible titles, and provide **citations** for all of the information included.
- Infographics consider **accessibility**: Make sure text is not cut off or covered by images, add descriptive alt-text to your infographic image file.

For more information on **accessibility**, please see: **DITI accessibility handout**

Feel free to ask questions at any point during the presentation!

Why Use Infographics?

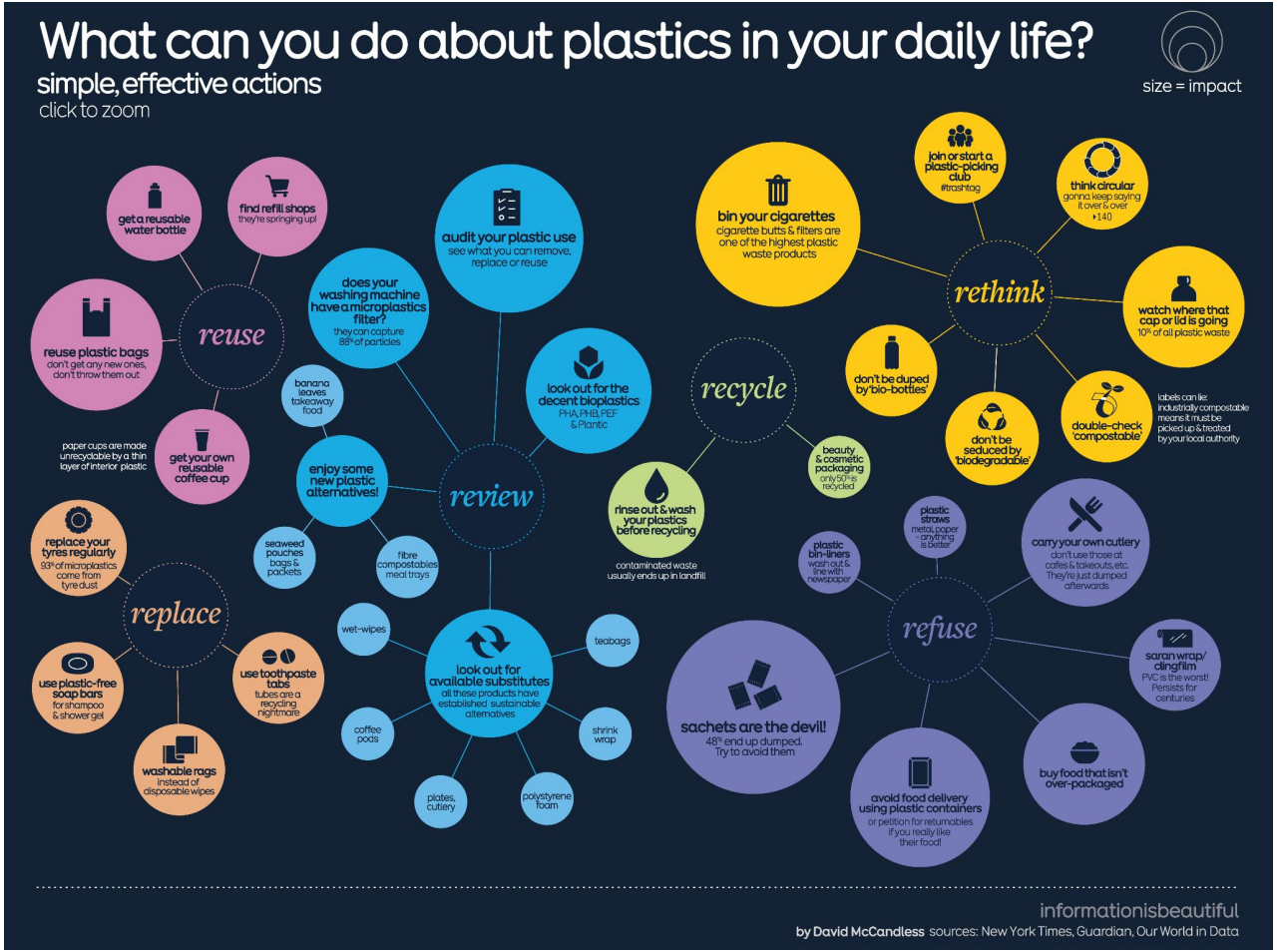
Infographics as a genre of communication have unique affordances and limitations.

Their concise nature and combined visual-verbal modes are great for quickly and efficiently communicating key information. However, as a result, infographics also provide you with very limited space, requiring you to think critically beforehand about what data you want to prioritize, what information is the most important to convey, etc.

Infographics Example 1

- What message/information is this infographic conveying? To whom?
- Why do you think the creator chose to communicate via infographic?

Source: "[What can you do about plastics in your daily life?](#)", David McCandless, [Information is Beautiful](#)



Infographics Example 2

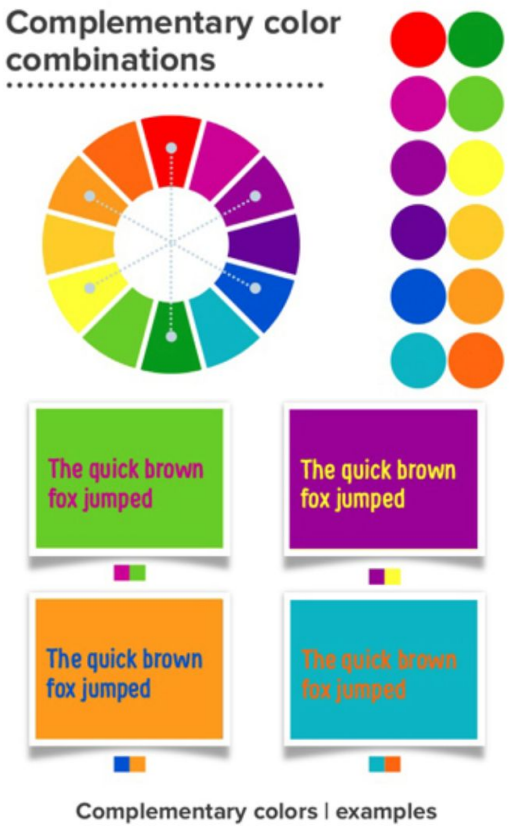
- Who is the audience for this infographic?
- How do you think that the visual style enhances (or potentially detracts from) the information?
- What similarities and/or differences do you see between this infographic and the previous?



Source: "BPL 2025 Snapshot," The Massachusetts Board of Library Commissioners (MBLC)

Keep it Visible: Colors

- Make color choices that are accessible to everyone (and appropriate for the presentation).
- Ideal color schemes are **high contrast** (use a contrast checker).
- Think about limiting your palette (3 colors can be effective).
- Find complementary colors (e.g. this tool).
- NU Colors provides the Northeastern University palette.



Visme Blog

Accurately Representing Data in Visuals and Infographics

—Developed in collaboration with BARI

Data Presentation Tips

- **Create your own tables**, or make sure to use only images that are shared with permissions that support reuse—and always cite your sources!
- **Be sure to present your data *accurately*—**be mindful that your charts, graphs, maps, and infographics are scaled and structured to present data and conclusions *completely* and *correctly*.
- **Use visual representations of numbers—**this will help concretize abstract concepts.
- **Label judiciously**, but don't overwhelm the viewer with dense text.
- **Beware of trying to make too many points in one graphic—**focus on the big takeaways.

Limitations of Some Data Presentation Methods: Charts, Graphs, Diagrams, Maps

- The **structure** and **scale** of charts and graphs could be **manipulated** to amplify or diminish differences.
- **Different types** of graphs and charts work better for some types of data than others—for example, a pie chart and a line graph might not both be able to represent the same data accurately.
- A chart with **too much information** will be difficult to understand, but **too little information** could be an indication that data has been cherry-picked to support an argument.
- There is **limited space** in an infographic for in-depth analysis; nuances can be flattened and obfuscated.

Misrepresentation of Data

From D.B. Resnik, in the *International Encyclopedia of the Social & Behavioral Sciences*, 2001: "The concept of 'misrepresentation,' unlike 'fabrication' and 'falsification,' is neither clear nor uncontroversial. Most scientists will agree that fabrication is making up data and falsification is changing data. **But what does it mean to *misrepresent* data? As a minimal answer to this question, one can define 'misrepresentation of data' as 'communicating honestly reported data in a deceptive manner.'"**

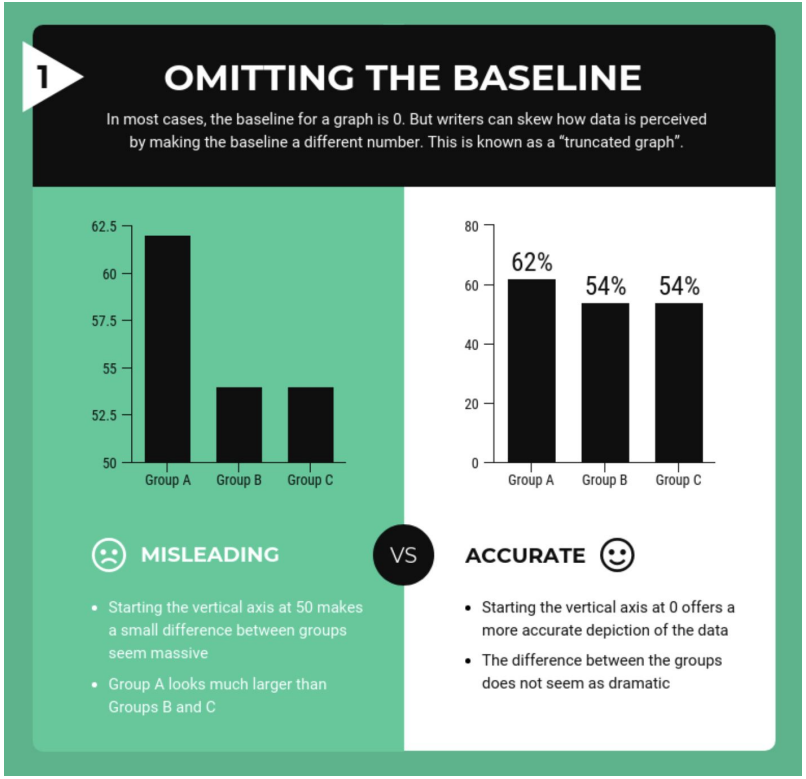
This **online book from The Data School** covers some common ways data could be misrepresented at multiple points in the process of gathering, analyzing, and presenting findings on data-based research.

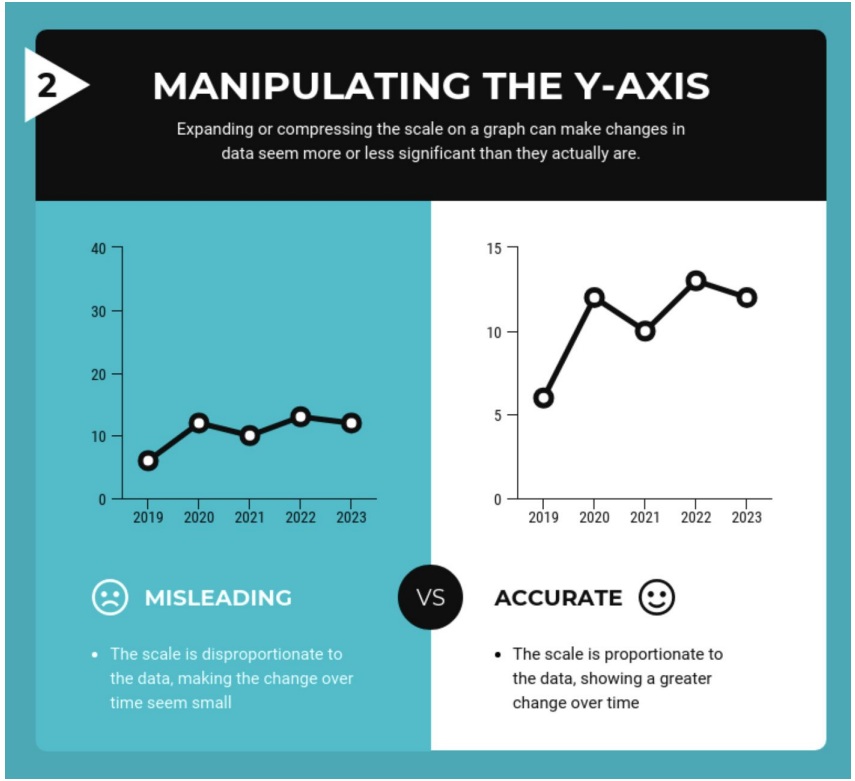
Limitations of Charts, Diagrams, Graphs, & Maps

Consider these questions as you review the examples in this section:

- What **commonalities** do you notice among the more misleading and more accurate versions of graphs and charts in these examples?
- How would you define "**accuracy**" in the context of data presentation? Why is that question essential to ask?
- In what **contexts** does it make the most sense to use these kinds of visuals to present data? Are there other times where they're inappropriate? How so?

Misleading axis representation



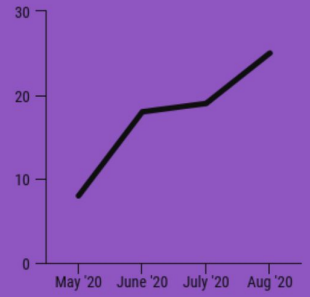


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Misleading data and graph selection

3 CHERRY PICKING DATA

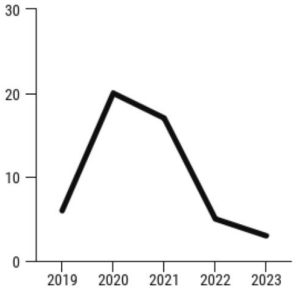
Writers may only include certain data points on their graphs to reinforce their narratives. This can create a false impression of the data.



MISLEADING

- Only a few months out of the year are graphed, depicting an upward trends

VS

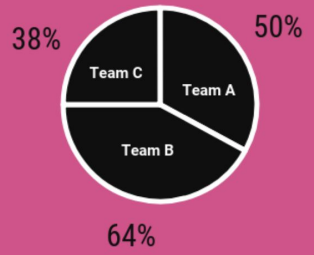


ACCURATE

- A much wider date range is graphed, revealing an overall downward trend
- This graphs shows the bigger picture

4 USING THE WRONG GRAPH

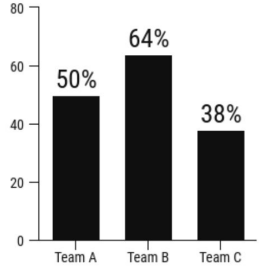
The type of graph you use should depend on the type of data you want to visualize. Using the wrong type of graph can skew the data. Writers will sometimes use the wrong type of graph on purpose.



MISLEADING

- Pie charts are used to compare parts of a whole, not the difference between groups
- A different type of graph should be used to compare the three teams

VS

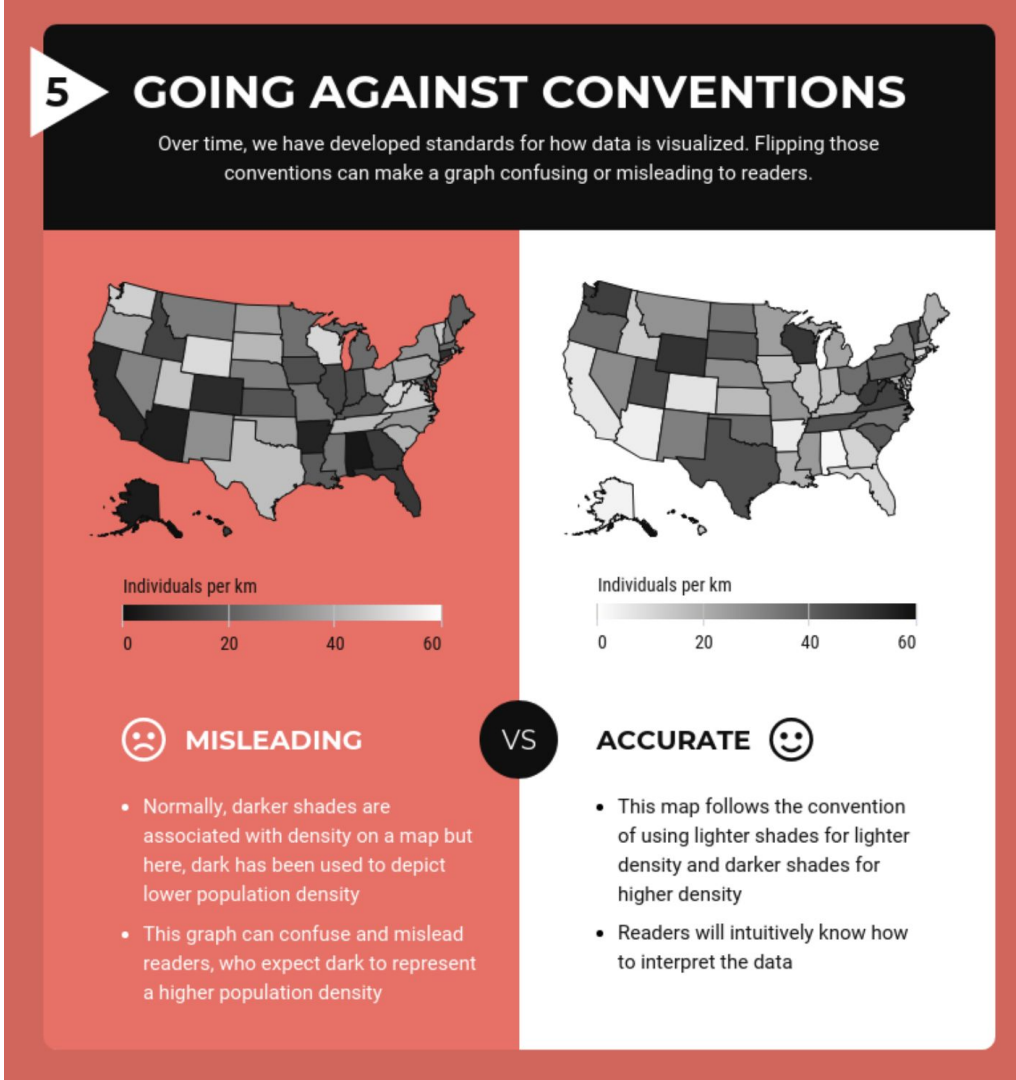


ACCURATE

- Bar graphs are better for showing the differences between groups
- This chart is a better visualization of the data

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Misleading map representations



Example of an opinion piece with visualizations

This example includes:

- A bar chart
- A map
- A line graph
- A table graph



Data gathering tips

Always remember your argument, and think about what you want your visualization to convey.

- Quantitative data
- Qualitative data
- Time (years, months, and days)
- Percentages (increase, decrease, and consistency)
- Geographic locations (countries, provinces, and cities)
- Track changes (trends)
- Organize your data in Excel, Google Sheets, or Canva Sheets

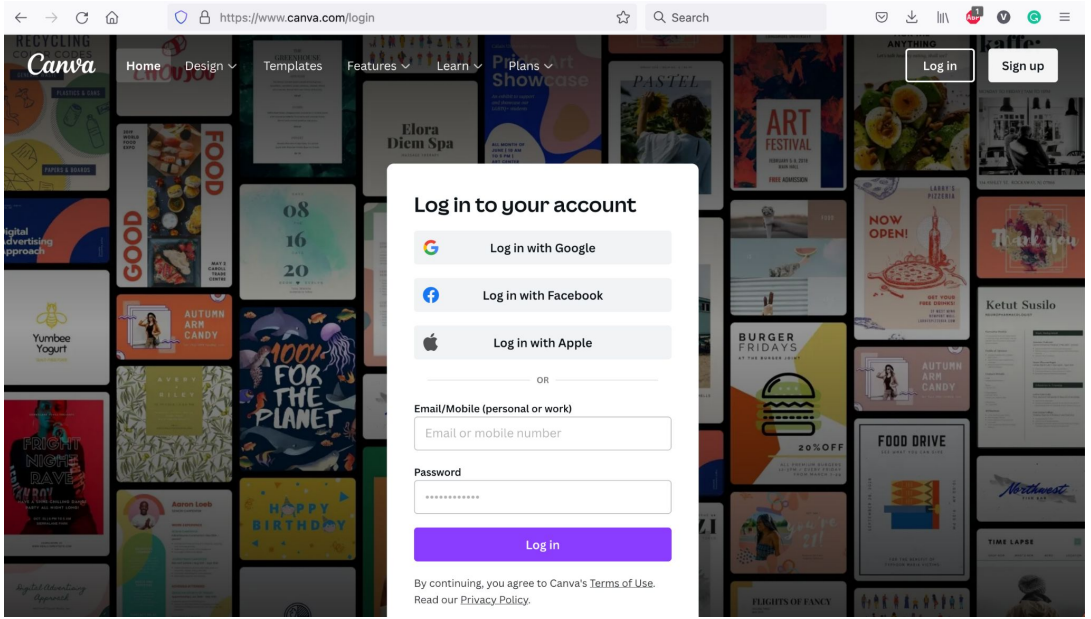
Getting Started with Canva

Data Privacy

- It's important to pay attention to data privacy when using digital resources
- At its simplest, **data privacy** is a person's ability to control what of their personal information is shared and with whom.
- To help you make informed decisions about interacting with digital tools in ways that honor your boundaries with your data and/or personal information, The DITI has prepared a handout on **Data Privacy**

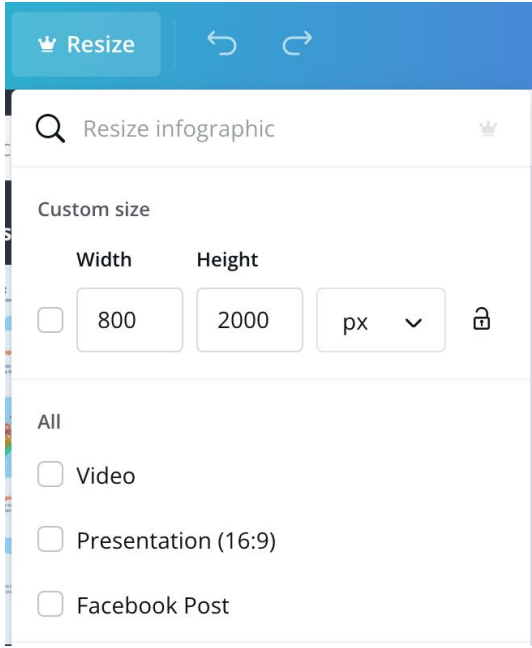
Sign up/Login

- Canva is a free online infographic maker.
- Use your Gmail/Facebook/Apple ID to sign up or create an account.
- Click “Create A Design.”



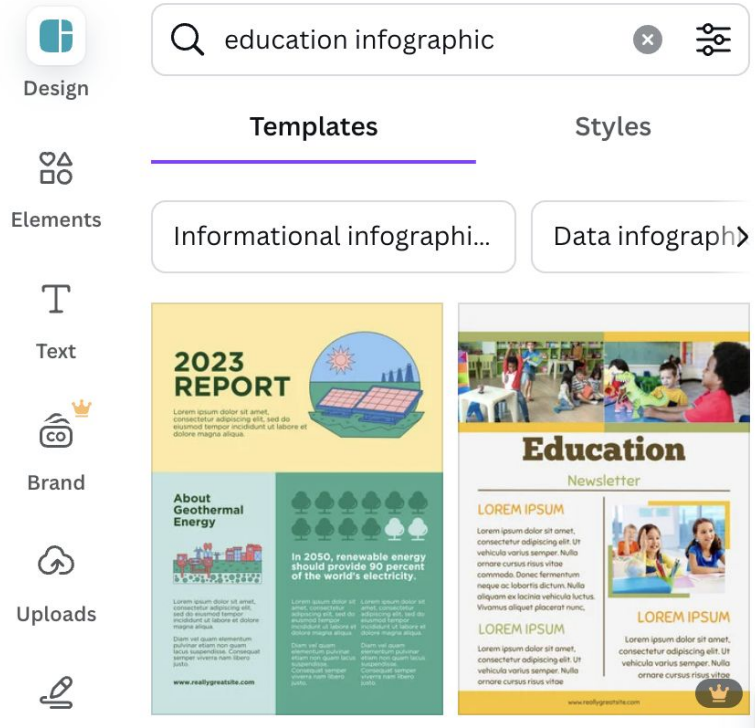
Starting Up...

- First, pick your canvas dimensions. Canva features a curated ‘infographic’ size, but you can customize the width and length when opening a new graphic. You can also add more ‘pages.’
    - Unfortunately, resizing and adding pages in an already existing graphic is only available in paid subscriptions.
- Infographics work best when presenting information through one direction, so it might help to size your canvas as being long either vertically or horizontally.



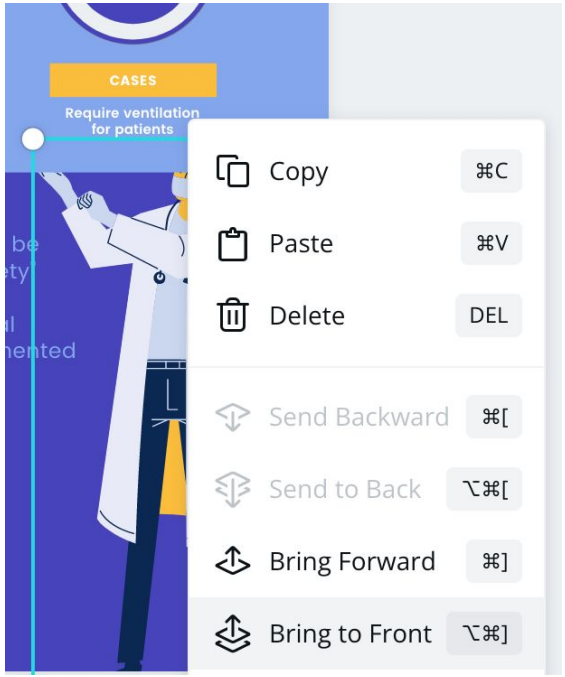
Pick a Style

- Infographics work best when presented in a uniform style.
- Try to get your information and your design to work together!
- If you need some help or inspiration, Canva offers several infographic templates you can start from.



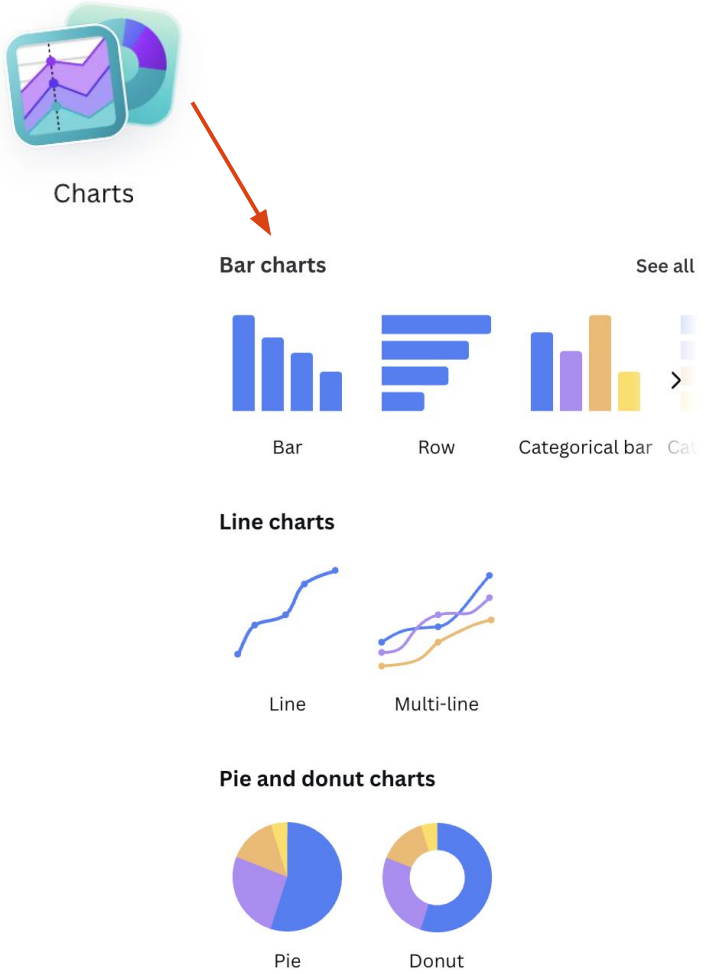
Layers

- Canva works by combining several graphic layers.
- Layers can be anything that shows up on the canvas—elements, charts, text, etc.—and can be customized in many ways. You can drag them around, change their colors, resize them, and more!
- Remember that you can always send a layer ‘forward’ or ‘back’ relative to other layers with a right click on PC, control click on Macs (or use the key commands shown at right) →



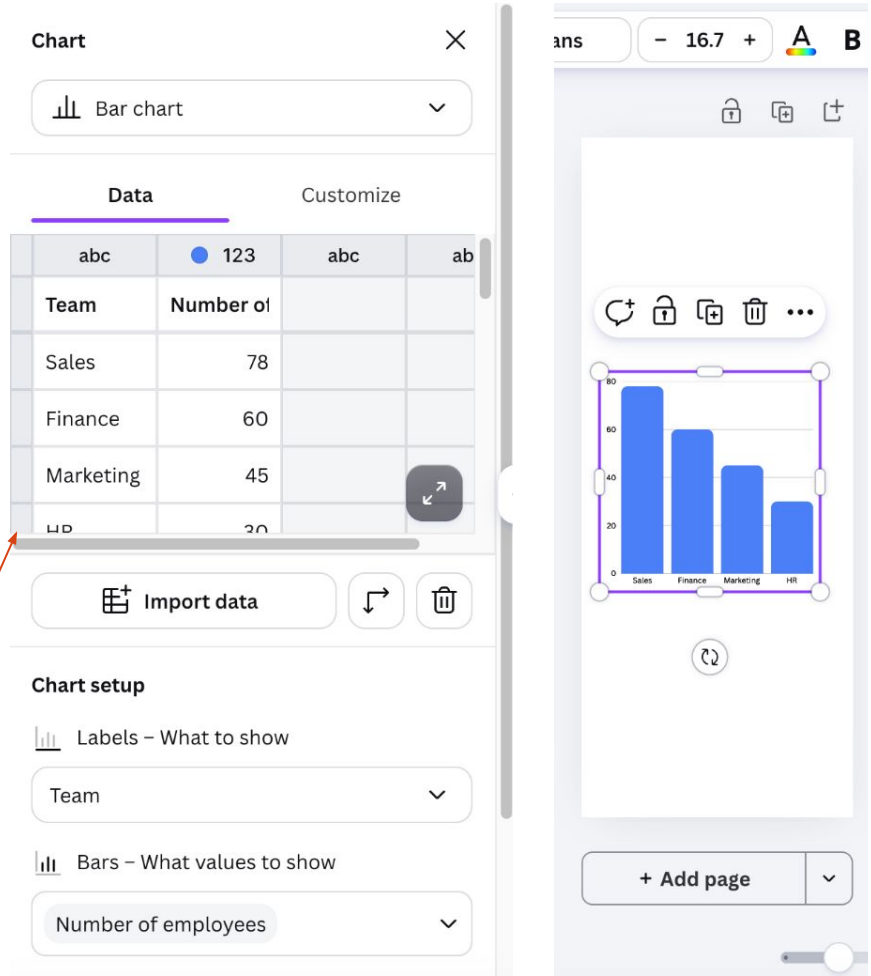
Creating Graphs 1/3

In the “elements” section, select “charts.” You will be presented with many different chart and graph options, including bar charts, line graphs, pie charts, area charts, scatter plots, etc.



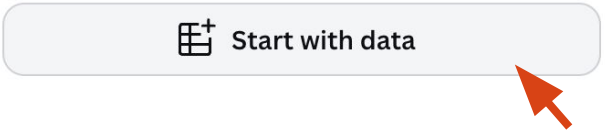
Creating Graphs 2/3

- If you already know which style graph you want to use, you can click directly on your selection, and a base graph will populate on your Canva page/project
- From there, you can edit the data in the auto-populated table that Canva creates for you, which will automatically adjust the visual graph based on the data you input

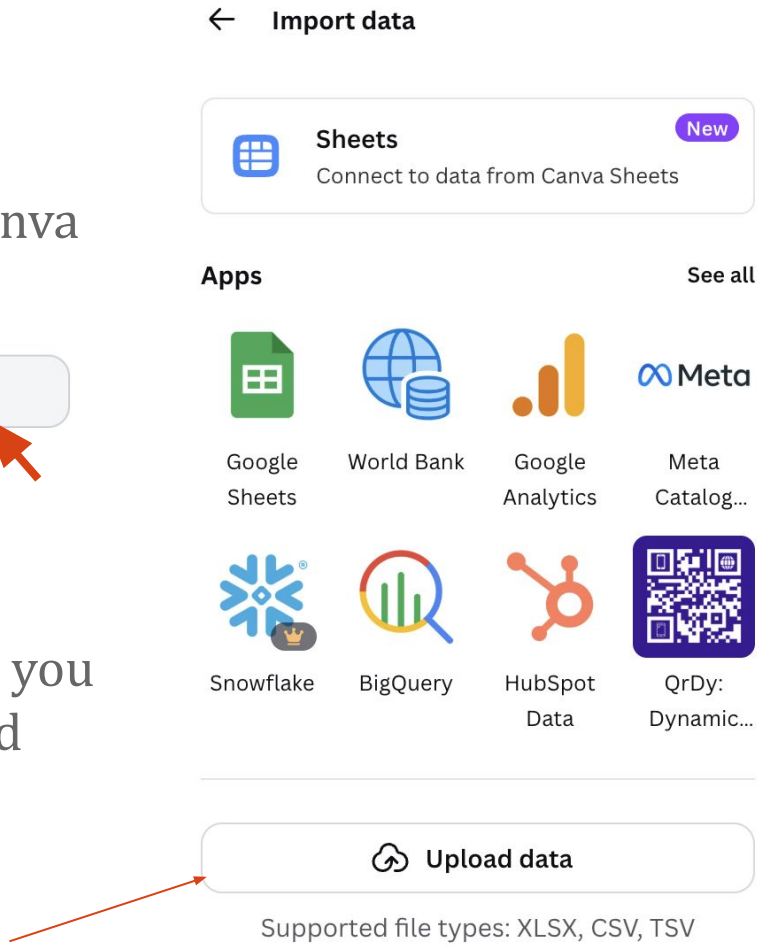


Creating Graphs 3/3

- You can also create charts or graphs in Canva by inputting your own data directly

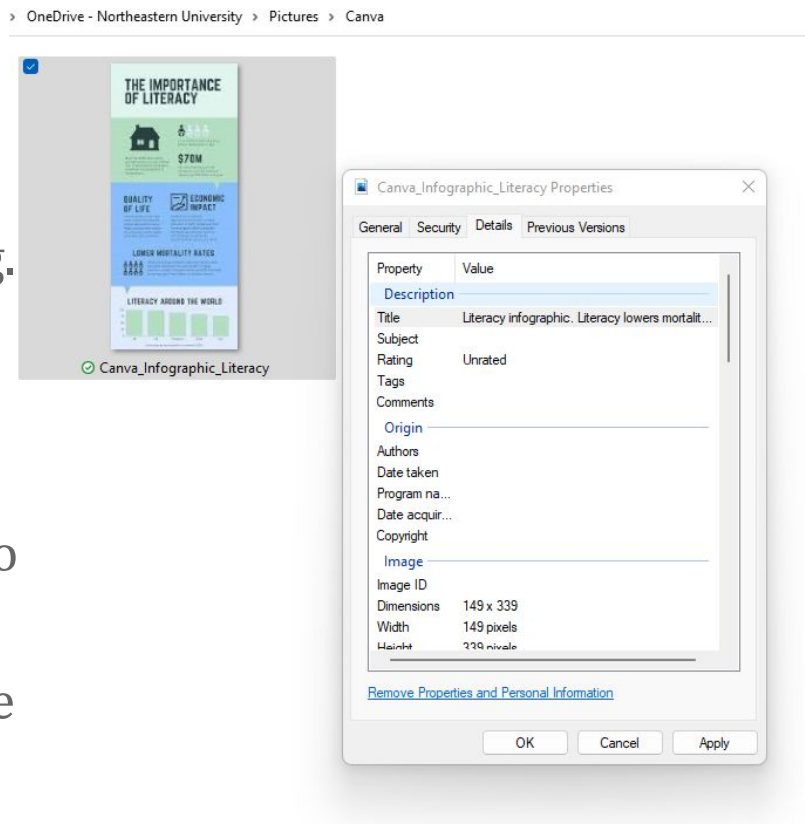


- Canva allows you to link data through a variety of other files and/or platforms, so you don't have to reproduce your data by hand



Alt-Text

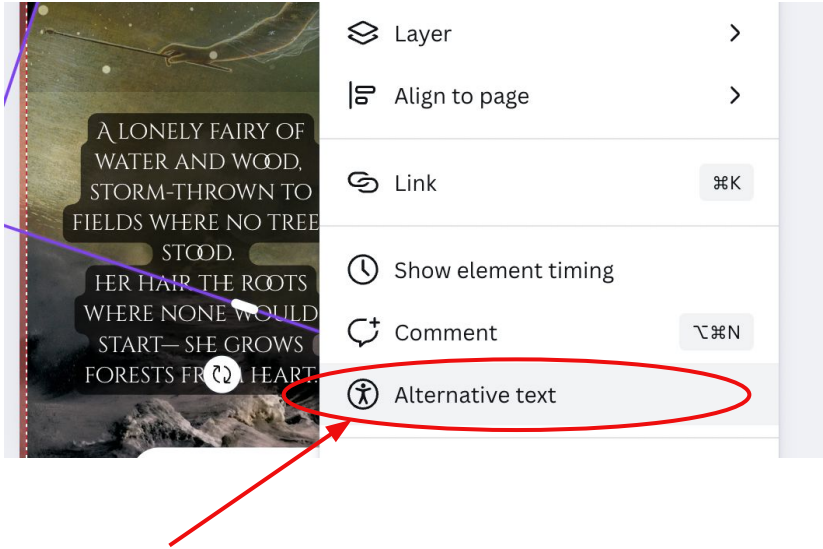
- Alt-text is a description of an image that conveys the image's content and meaning. This descriptive text is attached to the image's file properties.
- People with low or no vision can use assistive technology like screen readers to hear the alt-text description of digital images. When a screen reader reaches the image, it will read the alt-text description out loud.



Based on : [Microsoft's How to write effective alt-text]

Adding Alt-Text in Canva 1/2

In the Canva editing page, double-click on your image, and a menu will open to the right (the same menu as when you add layers). Click on the option for “Alternative text”

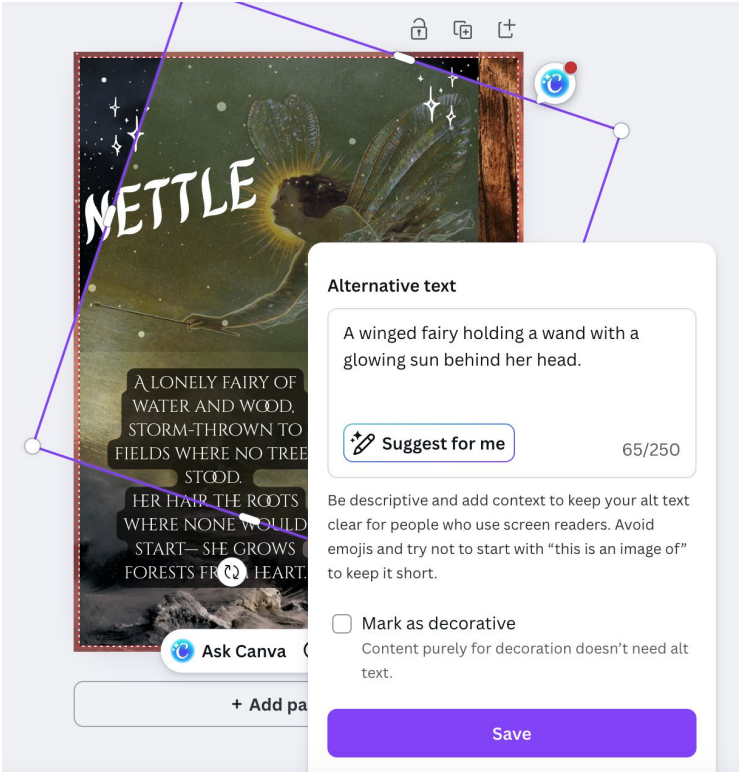


Feel free to ask questions at any point during the presentation!

Adding Alt-Text in Canva 2/2

A text bubble will appear where you can add the descriptive alt-text.

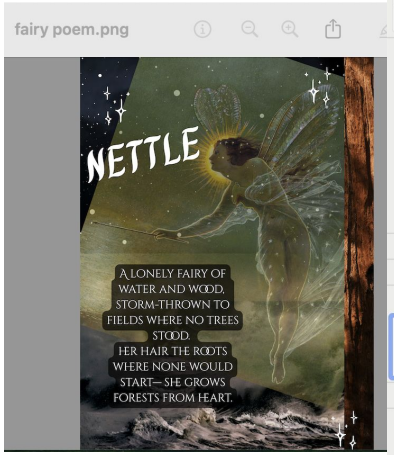
Note: this method means that you can add alt-text to each image separately, which is especially important if you are creating a file that contains multiple overlapping images or elements.

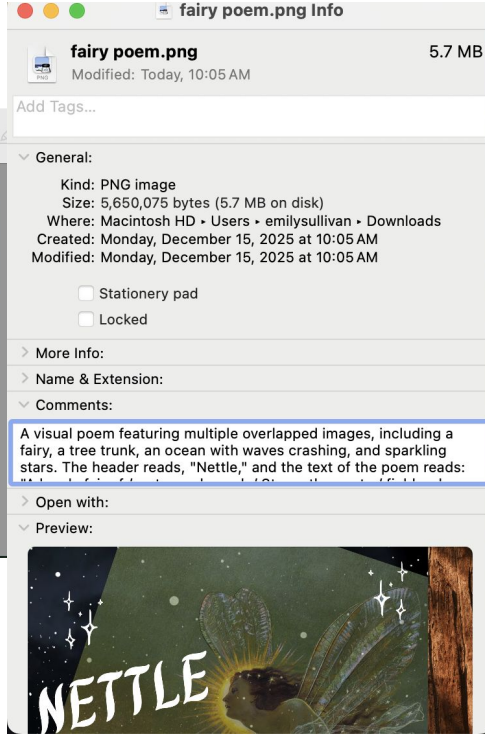


Feel free to ask questions at any point during the presentation!

Adding Alt-Text to a downloaded image file

- You can export your infographic as an image and then add alt-text.
- Find the image in your file directory.
- On PC: Right-click the image and select ‘Properties.’ Choose the ‘Details’ tab and click on the ‘Title’ field.
- On Mac: Control-click the image and select ‘Get Info’ on Macs, then edit the ‘Comments’ field.
- Write a description of and insights from your infographic.





Feel free to ask questions at any point during the presentation!

More Tips & Info to Consider

Build a coherent argument or narrative

- Treat your infographic like any other form of argument, explanation, or narrative—be **intentional**, and **organize** your points chronologically or as ordered steps in a process.
- Put **main ideas front-and-center**, and consider having your points progress down orders of importance.
- Use **signposts** or **sections** to orient your audience, i.e. pairing each point with numbers, shifting color gradients, etc.

Feel free to ask questions at any point during the presentation!

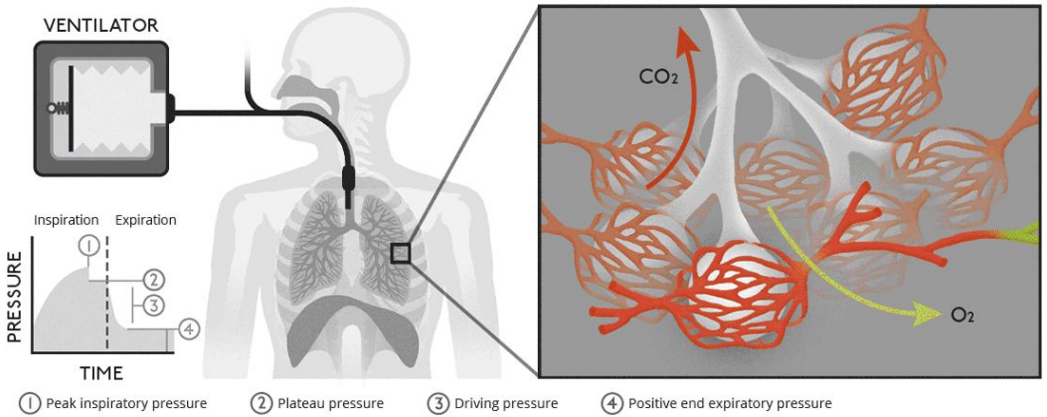
Some more tips

- Draw up an outline or storyboard **before** you build the graphic.
- Have a specific and **clear title**.
- Use **carefully-proofed syntax and vocabulary**, and explain terms/jargon.
- Have **proper citations**.
- Don’t mix too many visual types within your infographic. Keep a consistent font, color scheme, animation effects, design, formatting, etc.
- Limit text—keep negative space for the graphical component.

Feel free to ask questions at any point during the presentation!

Infographic Formats 1/2

- **Static** infographics:
  - Typically fixed information a still image.
- **Motion** infographics:
  - Typically fixed information. Display output is animated, or moving.



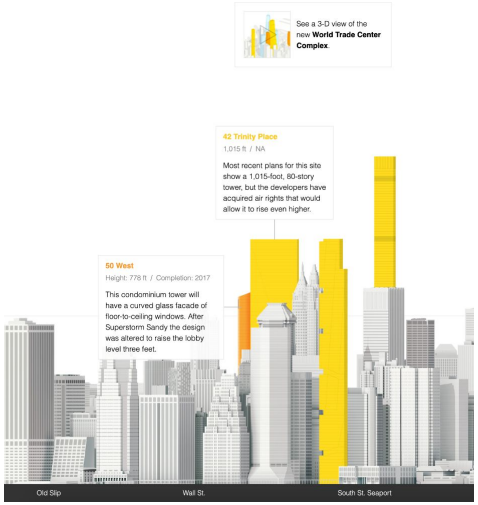
"Biology of ventilation," tapletopwhale.com

Feel free to ask questions at any point during the presentation!

Infographic Formats 2/2

Interactive infographics:

- Can be fixed or dynamic information input.
- User interaction consists of searching for specific data, actively shaping the content displayed, and choosing which information is accessed and visualized.

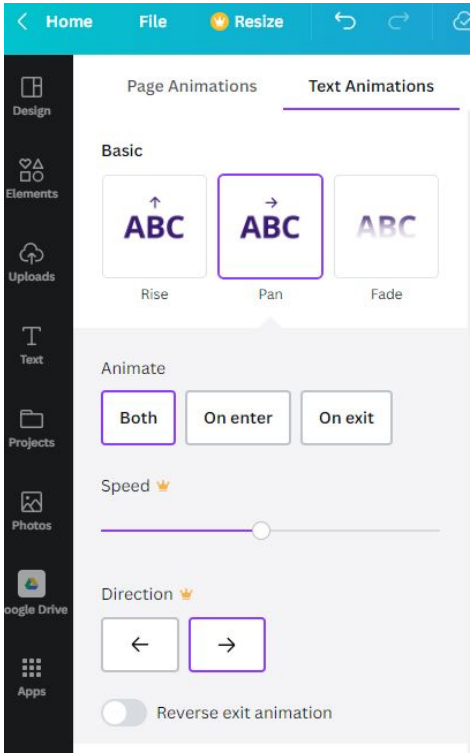


From "The New New York Skyline," *National Geographic*

Feel free to ask questions at any point during the presentation!

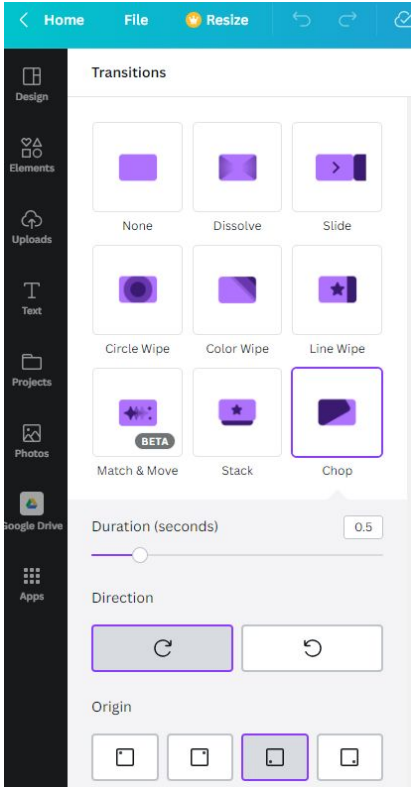
Canva Next Steps: Animations, Transitions, and Audio/Video

Adding Animation Effects



- There are multiple types of animations in Canva.
- To use animations, click on and select the feature you want to animate.
- Click on 'Animate' in the top menu, and review the types of Text Animations on the left-hand side of the screen. Hover your cursor over each type to preview how it looks.
- Page Animations apply the effect to all of the features on the slide.

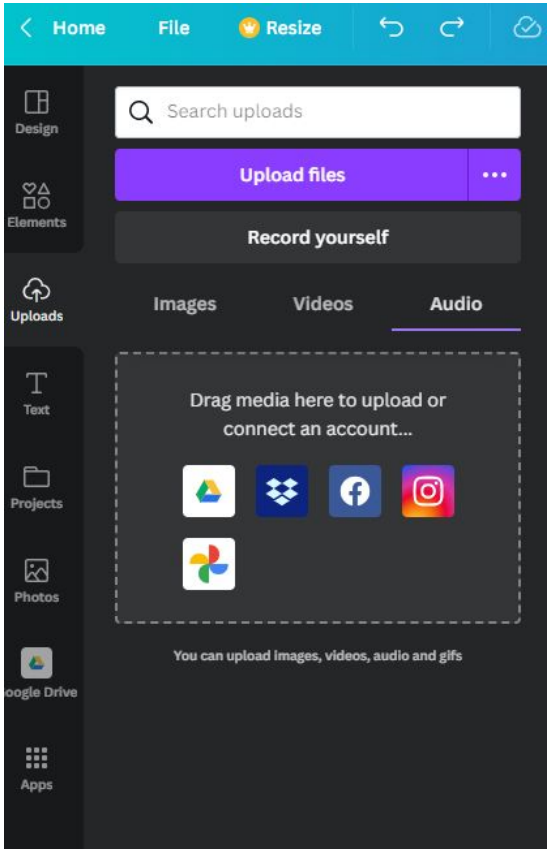
Adding Slide Transitions



- Slide transitions animate the movement from one slide to the next.
- To use transitions in Canva, click on the three dots that pop up above the slide in the preview deck at the bottom of the screen.
- Click ‘Add transition’ and review the options on the left side of the screen. Hover your cursor over each option to preview the transition effect.
- Once you choose a slide transition, Canva will let you adjust the duration and direction of the transition.

Use Audio/Video for Impact

- You can upload audio from your computer or record your own audio within Canva under the Uploads tab. Canva also has free audio and video you can use under the Elements tab.
- You can upload a video saved to your computer or link to an online video using the Canva Apps tab, which connects to YouTube and other platforms.
- Once you add in the audio or video to the Canva presentation, you can click on it to customize audio and video effects.



Canva & AI Tools

Using AI on Canva

Canva has a number of built-in AI tools and features. If you choose to use these features, you should first consider 1) your professor's policy on AI use and/or the guidelines around use of AI in your professional setting, 2) the technical, conceptual, and ethical implications of using generative AI, and 3) that any use of AI must be appropriately cited (see our handout on Copyright and Fair Use for some guidance on citing AI). You can also explore the DITI's page on AI for more of our resources on this topic.

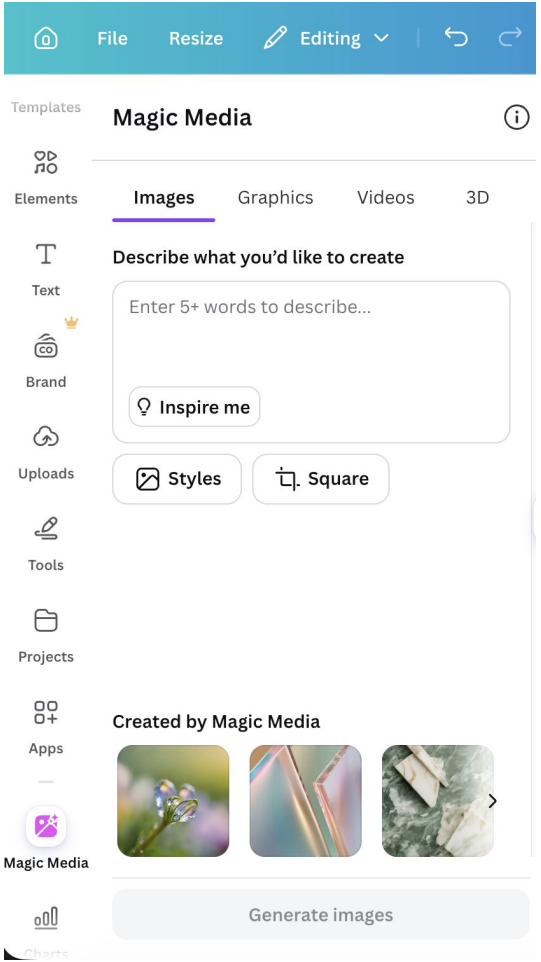
Also note that, as with many of Canva's tools, some are paid and some are free. For the purposes of this module, we will only cover the features that are usable with the free membership.

Canva's Magic Media

On Canva, you can use AI tools to:

1) edit existing media (change the visual style of an image into a variety of artistic forms, generate a short video from an image, etc.), or

2) use the generative AI feature to "create" visual media from a textual prompt.

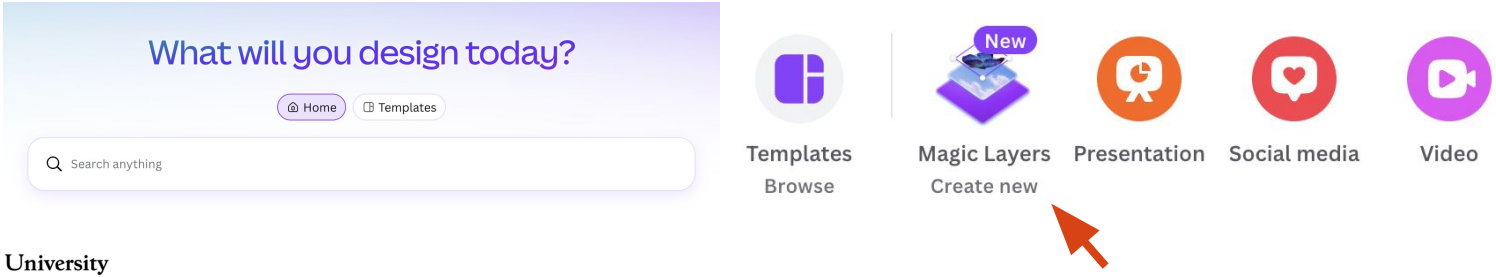


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Magic Layers 1/2

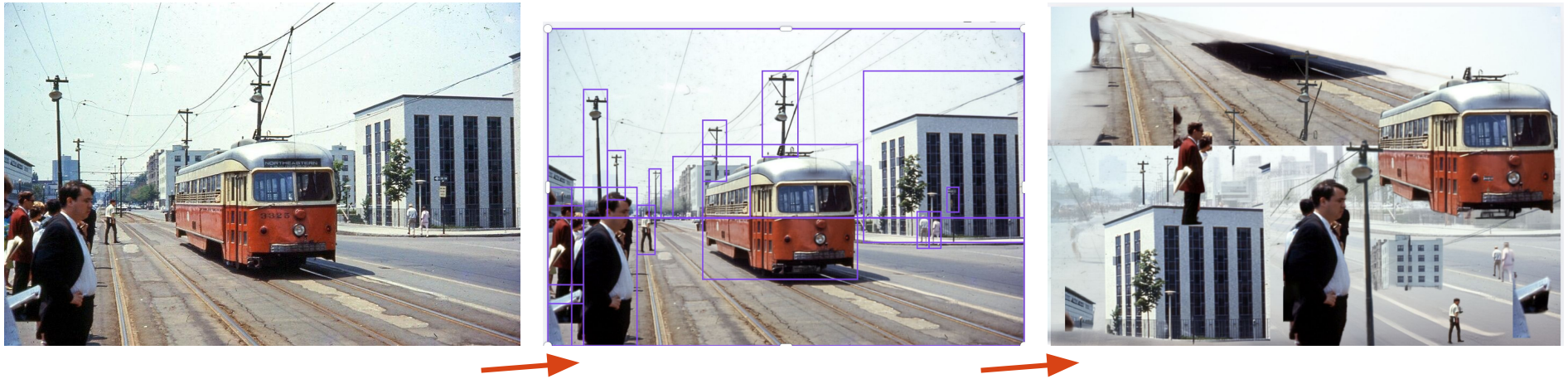
"Magic Layers" uses AI technology to digitally separate visual elements of an image, allowing you to extract specific visuals, move visuals around within a single photo, etc.

To use Canva's "Magic Layers" feature, **you have to start from the home page** (as opposed to editing your image directly in your project once you've already started). By pressing "Magic Layers," Canva will prompt you to upload your image.



Magic Layers 2/2

For example, here is a photo of the MBTA from Wikimedia Commons. Using the Magic Layers feature, the image of the train can be separated from the background, which you could then copy and paste onto your own project.



Source: David Wilson, CC BY 2.0 <https://creativecommons.org/licenses/by/2.0>, via Wikimedia Commons, https://commons.wikimedia.org/wiki/File:MBTA_3325_at_Northeastern_University_in_1967.jpg

Other Canva AI Tools



Learn Grid

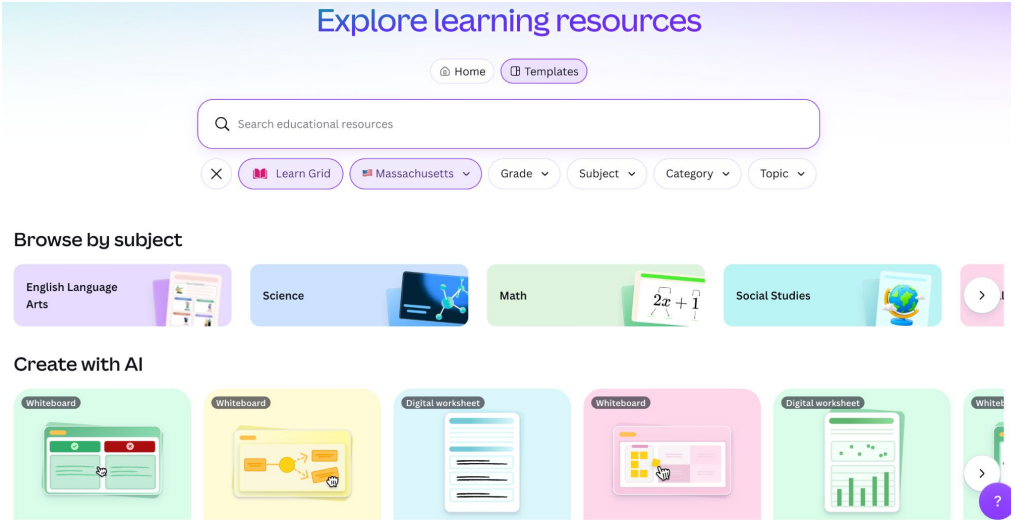
See all

Learn Grid a sub-page of Canva dedicated to teaching and learning resources, where you can use generative AI and textual prompts for new projects, or explore Canva's resources and templates

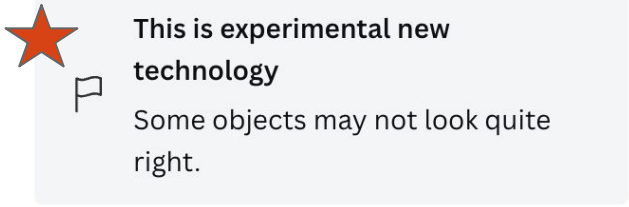


Code

Canva Code is a generative AI technology that allows you to create code based on textual prompts



AI Considerations 1/2



Technical

- There is a limited number of times you can use the free AI tools on Canva per month
- AI technology is experimental and can be unreliable (see Canva's disclaimer in the image above)★

Conceptual + Ethical

- Generative AI technology does cannot "create" independently; it draws from (often biased) training data comprising real peoples' work
- If you are creating an infographic using data, consider whether this is data that can ethically be shared with the AI tool (see our handouts on Data Privacy and Data Ethics for more on this topic)

AI Considerations 2/2

Before using AI features, ask yourself:

- Am I leveraging AI as a tool to accomplish my own creative and/or intellectual goals?
- Am I using AI to accomplish part of a task that I am still directing (e.g., using the “Magic Layers” editing feature to extract a single visual that you repurpose for your infographic), or am I using AI to do the entire task (e.g., prompting generative AI to make your infographic for you)?
- Am I using AI to accomplish something that I could do myself?
- How can I draft my prompt to be as precise as possible, tailoring expectations, instructions, and/or from what sources I want the AI tool to draw upon, with the least amount of prompting?

For Further Exploration

DITI infographics handout

DITI accessibility handout

DITI data privacy handout

DITI copyright and fair use handout

NULab Meet the Method: Canva

See also the Beginner's Guide and published by Canva

Feel free to ask questions at any point during the presentation!

Thank you!

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- Please fill out this survey to give us feedback: https://bit.ly/diti-feedback
- For more information on DITI, please see: https://bit.ly/diti-about
- Schedule an appointment with us! https://bit.ly/diti-meeting
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